Aditya R Dharamshi

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Location: Bengaluru, India (Ready to relocate)

Education

Linkoping University (Linköpings Universitet) — Linköping, Sweden
Masters of Science in Electronics Engineering (Analog and RF IC) — *2023 – 2025*
Focused on Analog CMOS, RFIC and Mixed Signal IC design, culminating in an 8-bit SAR ADC chip tape-out and lab evaluation project. Completed advanced coursework in RF systems, microwave engineering and antenna design.

Atria Institute of Technology — Bengaluru, India
Bachelors of Engineering in Electronics and Communication — *2016 – 2020*
Member of Center of Excellence, where I designed embedded hardware projects and participated in technical exhibition.

Projects

Design and Evaluation of Microstrip Resonator Based Dielectric Sensor - Master's Thesis — 2025

- Designed and experimentally validated a 1 GHz planar microstrip sensor with a dual split-ring resonator topology for dielectric characterization of liquids and materials.
- Conducted EM simulations using Keysight ADS Momentum Microwave and EMPro to optimize the geometry and analyze the S-parameters to maximize loaded Quality-factor (Q-factor).
- Fabricated the microstrip sensor on a custom PCB and implemented 3D-printed fixtures to conduct comparative material analysis and volume-dependence evaluations of different fluids.

8-bit SAR ADC (Taped-Out) – Mixed-Signal CMOS IC Design and Silicon Validation — 2024

- Designed and optimized the transmission gates for a bottom sampling differential capacitive DAC block in 350 nm CMOS technology in a multi-member design team.
- Developed high-level behavioral model using Verilog-A, performing full-custom mixed-signal transistor level schematic design, physical layout, padframe integration, and top-level routing within a $700\,\mu\text{m} \times 800\,\mu\text{m}$ area constraint (core area of $0.27\,\text{mm}^2$).
- Conducted pre-layout, post-layout and top-level chip simulations across all design phases, alongside physical verifications (DRC and LVS) during layout stages using Cadence Virtuoso (Spectre and ADE L).
- Conducted post-silicon package-level lab measurements of the 2 MHz SAR ADC using custom evaluation PCB to validate key performance parameters, including ENOB.

RF Transceiver - System level design — 2023-24

- Designed a 900-980 MHz direct-conversion RF transmitter subsystem for a TDD half-duplex transceiver, including architecture planning, SNR and link budget analysis and simulations using Keysight ADS.
- Analyzed transmitter performance parameters, including 8% EVM, ACPR requirements of -48 dBc and a 7 to 27 dBm output power range.
- Supported Zero-IF receiver design across the 1000-1080 MHz band, focusing on system level sensitivity (-96 dBm) and performance evaluations.

I^2S for Sound System using VHDL — 2023

- Designed and modelled an I^2S protocol interface block using Mentor Graphics HDL Designer, performing RTL simulations and timing analysis in ModelSim for a team-based sound system project.
- Designed and successfully validated keyboard and VGA modules using VHDL based on project specifications.

Professional Experiences

PCB Designer | Maxim Integrated (Analog Devices) | Bengaluru, India — Aug 2020 – Aug 2023

- Contributed to a company-wide PCB library audit project as an engineering intern, receiving corporate executive recognition before progressing to Associate PCB designer and subsequently to PCB designer.
- Designed evaluation, lab and ATE PCBs, implementing multi-layer stackup, routing, power distribution networks and layout constraints to mitigate EMI.
- Performed component placement, length matching, impedance control and signal integrity (SI) analysis.
- Coordinated with PCB fabrication vendors regarding DFM and manufacturing constraints, while and collaborating with global cross-functional engineering teams.

Electronics Hardware Intern (Low-Power Design) | EMickers AB | Sweden Feb 2026 – Mar 2026

- Designed a low-power IoT device with compact form factor for a sustainability-focused hardware project.
- Worked on schematics, component selection and HDI PCB layout using KiCAD, along with 3D CAD for board enclosure design.
- Coordinated with PCB fab vendors on design constraints and ensured manufacturable design, leading to successful first-pass board bring-up.

Volunteering Experience

Electronics Hardware Design Volunteer | SeeGoals, FIA Robotics July 2024 – July 2026

Linkoping University, Sweden

- Converted a wire-heavy football-playing robot system into a PCB-based design, improving reliability and stability.
- Designed a modular hardware system by separating critical functionality into replaceable daughterboards, improving fault isolation and maintainability
- Used basic STM32 firmware for hardware testing and validation of PCB prototypes.
- Integrated multiple subsystems into a single mainboard while working with firmware and mechanical teams to ensure system compatibility.
- Led the hardware design contributions that secured the team's qualification for RoboCup SSL 2026. This is recognized as a landmark qualification for Swedish representation.

Technical Skills

Core skills: Analog Mixed-Signal CMOS design, Custom IC layout, DRC/LVS/ERC, Top-level integration, Padframe routing, RTL design, timing analysis.

EDA and simulation tools: Cadence Virtuoso (Spectre, ADE L), Keysight ADS, EMPro, Mentor Graphics HDL Designer, ModelSim.

Silicon validation and lab: Post-silicon validation, Vector Network Analyzer (VNA), Spectrum Analyzer, Testbench Setup, Waveform Generators, Precision Soldering.

Hardware and PCB Design: High-speed PCB layout, Multi-layer stackup designs, Signal and Power Integrity (SI/PI), EMI/EMC mitigation, DFM/DFA verifications, Cadence Allegro PCB Designer, KiCAD.

Hardware Description and programming: Verilog, VHDL, MATLAB, C/C++, Python, Git.